- Arctic or polar cyclones occur in Antarctic regions and can reach up to 1,200 miles wide.
- Polar cyclones differ with others because they are not seasonal. [Tropical Cyclones are seasonal]
- They can occur at any time of the year.
- Polar cyclones can also form quickly (sometimes less than 24 hours), and their direction or movement cannot be predicted.
- They can last from a day up to several weeks. [Tropical Cyclones does not for more than a week]
- Most frequently, polar cyclones develop above northern Russia and Siberia.

93. **Statement 1: CORRECT**—Central dense overcast: This is the cirrus cloud shield that results from the thunderstorms in the eyewall of a tropical cyclone and its rain bands.
Statement 2: INCORRECT—The cyclones, which cross 20°N latitude generally, recurve and they are more destructive.
Statement 3: INCORRECT—In general, the strongest winds in a cyclone are found on the right side of the storm. The strongest wind on the right side of the storm is mainly due to the fact that the motion of the cyclone also contributes to its swirling winds.

94. **Polar Front Theory:**
- According to this theory, the warm-humid air masses from the tropics meet the dry-cold air masses from the poles and thus a polar front is formed as a surface of discontinuity.
- Such conditions occur over sub-tropical high, sub-polar low pressure belts and along the Tropopause.
- The cold air pushes the warm air upwards from underneath. Thus a void is created because of lessening of pressure. The surrounding air rushed in to occupy this void and coupled with the earth's rotation, a cyclone is formed which advances with the westerlies (Jet Streams).
- When the pressure drops along the front, the warm air moves northwards and the cold air move towards south setting in motion an anticlockwise cyclonic circulation (northern hemisphere). This is due to Coriolis Force.
- The cyclonic circulation leads to a well-developed extra tropical cyclone, with a warm front and a cold front.